Please make a copy of this document at t.ly/nV3J6

Complete the following forms from UNF to 3NF

A5-1.3 Multiple Forms Normalisation

Normalise the following forms describing a student/units system to 3NF. Remember you must show UNF, 1NF, 2NF, 3NF and amalgamate (do the attribute synthesis). Ensure you show all candidate keys at 1NF.

Note:

- To simplify a normalisation process that involves multiple forms, you **must** perform the normalisation one form at a time until all relations are in 3NF. Once you have done this process for all forms, consolidate the relations from the different forms (called attribute synthesis) by:
    - grouping together all relations with the same primary key, i.e. representing the same entity.
- choose a single name for synonyms. For example, mentor is the same as lecturer.

Normally we include all attributes in the form/report, but the report date in following reports is not an attribute that should be included in the normalisation process since it only represents the date when the report was printed. Also note that in this scenario, unit name and unit description are not unique.

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UNIT FORM

UNITS CURRENTLY APPROVED			REPORT DATE: 22/07/2022
Unit Number	**Unit Name**	**Unit Description**	**Unit Value**
FIT9131	Programming Foundations	Introduction to programming	6
FIT9132	Introduction to Databases	Database Fundamentals	6
FIT9134	Computer Architecture and Operating Systems	Fundamentals of computer systems and the computing environment	6
FIT9135	Data Communications	Fundamentals of data and computer communications	6

* Unit value may be either 3, 6 or 12 points

UNF (List all attributes and identify repeating groups)

UNIT(unit_no, unit_name, unit_desc, unit_value)

1NF (Remove repeating groups, identify PK for each relation and identify partial dependencies)

UNIT (unit_no, unit_name, unit_desc)
VALUE (unit_no, unit_value)

**Candidate keys*:

unit_no, unit_name
unit_no, unit_desc
unit_no, unit_value

Partial dependencies (a non-key attribute depends on only part of candidate key(s)).

Type here …

2NF (Remove partial dependencies and identify transitive dependencies)

Type here …

Transitive dependencies (a non-key attribute determines another non-key attribute/s).

Type here …

3NF (Remove transitive dependencies)

UNIT

Full dependencies

Type here …

LECTURER FORM

LECTURER DETAILS	**REPORT DATE:** 22/07/2022
LECTURER'S NUMBER: 10234 **LECTURER'S NAME:** GUISEPPE BLOGGS **LECTURER'S OFFICE No.:** 169 **LECTURER'S PHONE No.:** 99037111 **UNIT ADVISER FOR:**	
UNIT NUMBER	**UNIT NAME**
FIT9131	Programming Foundations
FIT9134	Computer Architecture and Operating Systems

* A given unit may have several advisers
* Some lecturers share offices, although each has their own phone

UNF (List all attributes and identify repeating groups)

LECTURER (lec_no, lec_name, lec_office, lec_phone, (unit_no, unit_name))

1NF (Remove repeating groups, identify PK for each relation and identify partial dependencies)

LECTURER (lec_no, lec_name, lec_office, lec_phone)

**Candidate keys*:
lec_no
lec_phone

ADVICE (lec_no, unit_no, unit_name)

**Candidate keys*:
lec_no, unit_no

Partial dependencies (a non-key attribute depends on only part of candidate key(s)).

unit_no -> unit_name

2NF (Remove partial dependencies and identify transitive dependencies)

LECTURER (lec_no, lec_name, lec_office, lec_phone)
ADVICE (lec_no, unit_no)
UNIT (unit_no, unit_name)

Transitive dependencies (a non-key attribute determines another non-key attribute/s).

None

3NF (Remove transitive dependencies)

LECTURER (lec_no, lec_name, lec_office, lec_phone)
ADVICE (lec_no, unit_no)
UNIT (unit_no, unit_name)

Full dependencies

lec_no -> lec_name, lec_office, lec_phone
lec_no- > unit_no
unit_no -> unit_name

STUDENT FORM

STUDENT DETAILS	REPORT DATE: 22/07/2022

STUDENT No.: 12345678
STUDENT NAME: Poindexter Jones
STUDENT ADDRESS: 23 Wide Road, Caulfield, 3162
COURSE ENROLLED: MIT
MODE OF STUDY: On-Campus

MENTOR NUMBER: 10234
MENTOR NAME: Guiseppe Bloggs

ACADEMIC RECORD:

UNIT NUMBER	UNIT NAME	YEAR / SEMESTER	GRADE
FIT9131	Programming Foundations	2021/2	N
FIT9131	Programming Foundations	2022/1	D
FIT9132	Introduction to Databases	2022/1	D

* Grade may have the value N, P, C, D or HD
* Mode of Study must be On-campus (O) or Distance Education (D)

In order to add a student, the lecturer who advises that student must already exist in the database. No lecturer who advises any students may be deleted from the database. If the lecturer number of a lecturer is changed, then the number would be changed for each student advised by that lecturer.

UNF (List all attributes and identify repeating groups)

STUDENT (stu_no, stu_name, stu_address, stu_crse, stu_mode, lec_no, lec_name (unit_no, unit_name, ar_year, ar_semester, ar_grade))

1NF (Remove repeating groups, identify PK for each relation and identify partial dependencies)

STUDENT (stu_no, stu_name, stu_address, stu_crse, stu_mode, lec_no, lec_name)

**Candidate keys*:
stu_no

AC_REC (stu_no, unit_no, ar_year, ar_semester, unit_name, ar_grade)

**Candidate keys*:
stu_no, unit_no, ar_year, ar_semester

Partial dependencies (a non-key attribute depends on only part of candidate key(s)).

unit_no -> unit_name

2NF (Remove partial dependencies and identify transitive dependencies)

STUDENT (stu_no, stu_name, stu_address, stu_crse, stu_mode, lec_no, lec_name)
AC_REC (stu_no, unit_no, ar_year, ar_semester, ar_grade)
UNIT (unit_no, unit_name)

Transitive dependencies (a non-key attribute determines another non-key attribute/s).

Type here …

3NF (Remove transitive dependencies)

STUDENT (stu_no, stu_name, stu_address, stu_crse, stu_mode, lec_no, lec_name)
AC_REC (stu_no, unit_no, ar_year, ar_semester, ar_grade)
UNIT (unit_no, unit_name)
LECTURER (lec_no, lec_name)

Full dependencies

Type here …

Attributes synthesis

UNIT (unit_no, unit_name, unit_desc, unit_value)
LECTURER (lec_no, lec_name, lec_office, lec_phone)
ADVICE (lec_no, unit_no)
STUDENT(stu_no, stu_name, stu_address, stu_crse, stu_mode, lec_no)
AC_RECORD (stu_no, unit_no, ar_year, ar_semester, ar_grade)

A5-2 Further Multiple Forms Normalisations

A5-2.1 Property Maintenance Forms

Create a new document called **A5-2** to write your answers for this activity.

Please note these forms are a continuation of the case you modelled under lesson A3-4 (you should include those case details as part of your considerations as you complete this task)

Normalise the following forms to 3NF. Remember, you must show UNF, 1NF, 2NF, and 3NF and amalgamate (consolidate/attribute synthesis). To simplify a normalisation process that involves multiple forms, you should perform the normalisation one form at a time until all relations are in 3NF. Once you have done this process for all forms, consolidate the relations from the different forms (called attribute synthesis) by grouping together all relations with the same primary key, i.e., representing the same entity.

> **When working with a potentially composite attribute, you must consider how it is depicted on the form/report**. If the attribute is depicted as non-decomposed, then you should treat it as a simple attribute for the purposes of normalisation. *However*, this decision must be informed by any further information that you have about the scenario you are modelling.

For example, your client has informed you that they record a tenant's given name and family name, whereas a form you are normalising has an entry such as:

Tenant Name: Johanston Karkov

In such a situation, this output is simply the manner which has been chosen to display the name on the form. As part of normalising, such an attribute must be decomposed into a given name and family name since you were informed that this is the way it is recorded.

The agency records normal property maintenance. Maintenance costs are charged to the property owner. There is only a single maintenance for each property at a particular date and time. Below is a sample of property maintenance report prepared for the property owner:

PROPERTY MAINTAINANCE REPORT

PROPERTY MAINTENANCE REPORT

Property No: 1965
Property Address: Unit 1 100-102 Platypus St, Oakleigh, VIC, 3166

Owner No: 9321
Owner:
  Given Name: Lilian
  Family Name: Potter
Owner Address: 14 Puffin Rd, Wantirna, VIC, 3152

MAINTENANCE RECORD:

DateTime	Description	Cost
2022-06-01 9:00	Roof leak	$854
2022-06-10 10:00	Lawn repair	$230
2022-06-10 14:00	Minor crack repair	$428

UNF (List all attributes and identify repeating groups)

Type here …

1NF (Remove repeating groups, identify PK for each relation and identify partial dependencies)

Type here …

**Candidate keys*:

Type here …

Partial dependencies (a non-key attribute depends on only part of candidate key(s)).

Type here …

2NF (Remove partial dependencies and identify transitive dependencies)

Type here …

Transitive dependencies (a non-key attribute determines another non-key attribute/s).

Type here …

3NF (Remove transitive dependencies)

Type here …

Full dependencies

Type here …

The agency also provides two samples of simplified property tenant ledgers (tenant payment history):

PROPERTY TENANT LEDGER (Sample 1)

Property No: 1965
Property Address: Unit 1 100-102 Platypus St, Oakleigh, VIC, 3166
Lease Start Date: 2021-01-14
Weekly Rental Rate: $425
Bond: $1,842

Tenant No: 512736
Tenant Name: Johanston Karkov

PAYMENT RECORD:

Number	Date	Type	Amount	Paid By
80010	2021-01-10	Bond	$1,842	bank cheque
80024	2021-01-10	Rental	$1,842	direct deposit
80501	2021-02-10	Rental	$1,842	bpay
80612	2021-02-28	Damage	$523	credit card
80615	2021-02-28	Damage	$312	credit card
80892	2021-03-12	Rental	$1,842	bpay
81073	2021-04-13	Rental	$1,842	credit card

PROPERTY TENANT LEDGER (Sample 2)

Property No: 1965
Property Address: Unit 1 100-102 Platypus St, Oakleigh, VIC, 3166
Lease Start Date: 2022-05-05
Weekly Rental Rate: $475
Bond: $2,059

Tenant No: 623124
Tenant Name: Kiky Longbottom

PAYMENT RECORD:

Number	Date	Type	Amount	Paid By
101829	2022-05-01	Bond	$2,059	bank cheque
101934	2022-05-03	Rental	$2,059	credit card
102104	2022-06-04	Rental	$2,059	credit card

Note that the two property tenant ledgers shown are two examples of the same form, only one normalisation process that covers both samples is required.

PROPERTY TENANT LEDGER FORM

UNF (List all attributes and identify repeating groups)

Type here …

1NF (Remove repeating groups, identify PK for each relation and identify partial dependencies)

Type here …

**Candidate keys*:

Type here …

Partial dependencies (a non-key attribute depends on only part of candidate key(s)).

Type here …

2NF (Remove partial dependencies and identify transitive dependencies)

Type here …

Transitive dependencies (a non-key attribute determines another non-key attribute/s).

Type here …

3NF (Remove transitive dependencies)

Type here …

Full dependencies

Type here …

Attributes synthesis